

## Data Flow Diagram

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|----------------------|---------------------------------|
| <b>Date</b>          | 26 June 2026                    |
| <b>Team ID</b>       | SWTID-2026-5060                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 2 Marks                         |

### Data Flow Diagram

Create a Data Flow Diagram (DFD) to illustrate how data flows through the Credit Card Approval Prediction System between external entities, system processes, and data stores. The diagram represents how applicant information is collected, preprocessed, passed to the machine learning model for prediction, stored in the prediction log database, and finally displayed to the user as a credit card approval decision along with a detailed risk audit report. Use the standard DFD symbols to represent external entities, processes, data flows, and data stores.

### DFD Symbol Legend

| Symbol                                   | Name                   | Description                                                                                                                                                                                            |
|------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oval / Rounded shape</b>              | <b>External Entity</b> | A person or system outside the application that provides or receives data (e.g., Credit Card Applicant, Bank Officer).                                                                                 |
| <b>Rectangle with numbered header</b>    | <b>Process</b>         | An activity that processes applicant information, performs preprocessing, executes the machine learning model, and generates the prediction (e.g., Validate Input, Preprocess Data, Predict Approval). |
| <b>Rectangle (solid fill, no number)</b> | <b>Data Store</b>      | A storage location where application data or prediction history is maintained (e.g., Prediction Log Database, Trained Random Forest Model).                                                            |
| <b>Labeled arrow</b>                     | <b>Data Flow</b>       | Represents the movement of applicant data, processed features, prediction results, and audit reports between external entities, processes, and data stores.                                            |

## Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



### CREDIT CARD APPROVAL PREDICTION – WORKFLOW (AFTER DEVELOPMENT)

*End-to-End Workflow from User Input to Final Prediction*

